

Task 1

List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

1. CMOS Checksum Error Defaults Loaded

This message means that the BIOS/CMOS configuration data does not match the expected checksum, so the firmware has loaded default settings. Common causes include a weak CMOS battery, a cleared/reset CMOS, or corrupted configuration data. The first checks are the CMOS battery, date/time, and BIOS settings.

2. Keyboard Error or Keyboard Not Detected

The POST has not detected a usable keyboard. The cause may be a disconnected keyboard, a faulty USB/PS2 connection, a damaged keyboard, or a motherboard USB/controller issue. Trying a known-good keyboard and another port is a basic first check.

3. CMOS Battery Low

This warning indicates that the battery used to retain CMOS settings is becoming weak. Typical symptoms include the system clock resetting and BIOS settings being lost after power is removed. Replacing the compatible CMOS battery and then saving the correct BIOS settings normally resolves it.

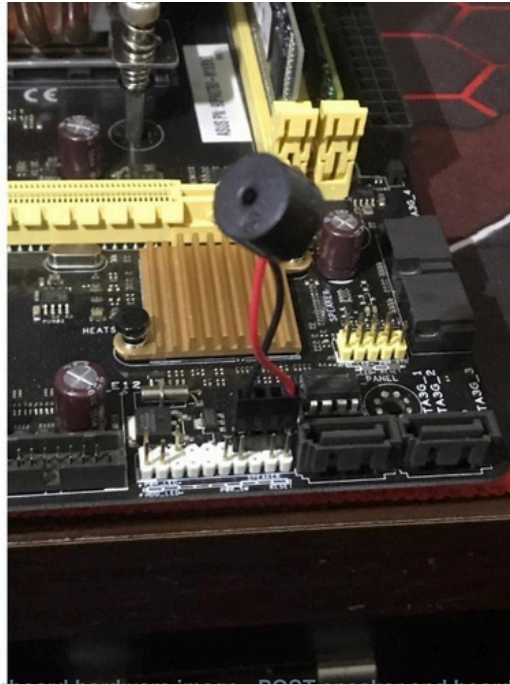
4. CPU Fan Error

The BIOS detects that the CPU cooling fan is not reporting the expected speed. A loose fan connector, blocked fan, failed fan or incorrect fan header connection can cause the warning. The CPU cooling system should be checked before continuing to use the PC under load.

5. No Boot Device Found / Operating System Not Found

The firmware cannot find a storage device containing a bootable operating system. Possible causes include a disconnected/failing SSD or HDD, incorrect boot order, a disabled storage controller, or a damaged boot configuration. Checking storage detection in BIOS and the boot order is an appropriate first step.

Task 2



Real motherboard hardware image - POST speaker and board components

Real motherboard hardware image - POST speaker and board components

Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

Meaning of the 3-beep pattern

For the classic AMI BIOS beep-code table, 3 short beeps indicate a base-memory read/write test error, which points toward a memory/RAM problem. However, beep patterns can vary by motherboard and firmware revision, so the motherboard manual should be checked before treating the code as a universal rule. cite

Troubleshooting Step 1 Reseat the RAM

Power the PC off and disconnect AC power. Remove the RAM module(s), inspect the slot and contacts, then firmly reinstall the module. If more than one module is installed, testing one known-good module at a time can help isolate a faulty module or slot. AMI documentation specifically recommends reseating the memory or replacing it with known-good modules for 13 beep memory faults. cite

Troubleshooting Step 2 Test the modules and slots

If reseating does not help, test the RAM modules individually and, where appropriate, try another compatible slot. This helps distinguish a bad RAM module from a slot/contact or motherboard problem. If the system still produces the same code with known-good memory, the motherboard or memory-controller path may require further diagnosis.

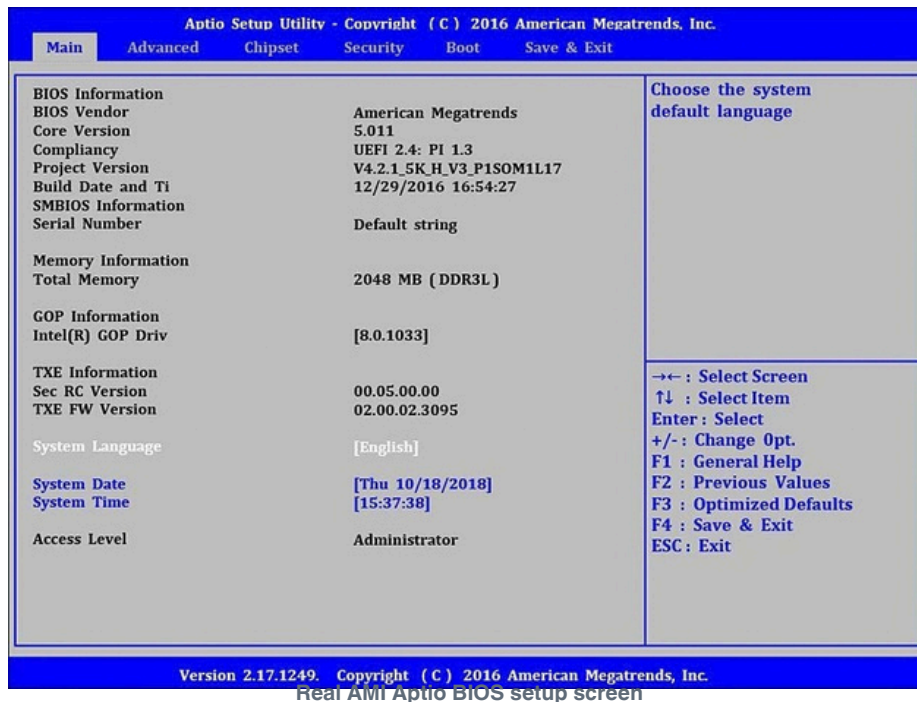
Real hardware image

A motherboard speaker/buzzer is the component that produces the POST diagnostic beeps. A real motherboard speaker example can be viewed from the web image reference linked below.

Real web image reference: Motherboard speaker/buzzer photo

https://mall.iopenmall.tw/website/uploads_product/website_28358/P2835804147608_4_36491962.jpg

Task 3



Real AMI Aptio BIOS setup screen

Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.

What to record from the BIOS Main screen

The BIOS Main/Information screen normally shows the BIOS vendor or manufacturer, BIOS/core version, build date, processor information, memory information and sometimes the motherboard/project identifier. For this assignment, write the values exactly as they appear on the system being used rather than copying the values from an example image.

Example of a real BIOS screen

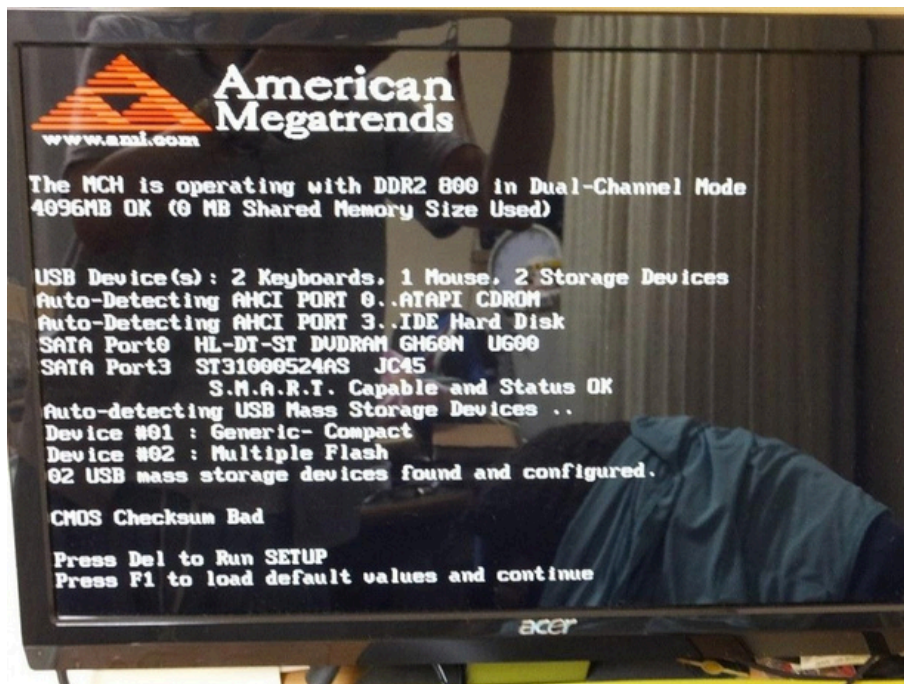
A real AMI Aptio Setup Utility Main screen shows the BIOS Vendor as American Megatrends and includes BIOS information, processor information, memory information and system date/time. The example is useful for identifying where the required fields are located. cite

How to complete the task

Restart the computer and enter BIOS/UEFI Setup using the key shown during startup, commonly Delete or a function key depending on the manufacturer. On the Main or Information page, record the motherboard/BIOS vendor and BIOS version. Then take a screenshot and attach it to the assignment. Do not include serial numbers or other personal/device identifiers if the submission does not require them.

Real web image reference: AMI Aptio BIOS Main screen https://www2.mouse-jp.co.jp/ssl/user_support2/faq/30943/01.jpg

Task 4



Real CMOS Checksum Bad POST screen

Real CMOS Checksum Bad POST screen

You are given a screenshot of a POST error message that says 'CMOS checksum error Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

What the message means

The BIOS has detected that the stored CMOS configuration data does not match the checksum it expects. As a protective measure, default BIOS settings are loaded so the system can continue with a known configuration. A real AMI BIOS screen can show messages such as 'CMOS Checksum Bad' and 'Press F1 to Run SETUP / load default values.'

Step 1 Check the date and time

Enter BIOS Setup and check whether the system date and time are incorrect or repeatedly reset after the PC is powered off. Repeated loss of date/time is a strong clue that the CMOS battery may be weak.

Step 2 Load safe/default BIOS settings

If the system boots after the warning, enter BIOS Setup and load the recommended default/optimized settings. Confirm that the storage mode, boot device and other required configuration values are correct, then save and restart.

Step 3 Replace the CMOS battery if required

If the checksum warning and lost date/time return after shutdown, replace the compatible CMOS battery, commonly a CR2032 on desktop motherboards. Follow the motherboard manufacturer's procedure and disconnect power before handling the battery.

Step 4 Check for repeated configuration loss

After replacing the battery, enter BIOS again, set the correct date/time and required boot settings, save the changes, and restart the computer several times. If the checksum error continues despite a good battery, investigate CMOS/firmware corruption, a reset jumper, motherboard faults or a BIOS update/recovery requirement.

Conclusion

The message does not automatically mean that the motherboard is permanently damaged. It indicates a problem with the stored BIOS/CMOS configuration and should be diagnosed by checking the battery, settings and any reason the CMOS data may be getting reset.

Real web image reference: AMI BIOS CMOS Checksum Bad screen
https://bbsimg03.kakaku.k-img.com/images/smartphone/icv/1994977_f.jpg

Real image references used for study: AMI CMOS checksum screen and a real motherboard BIOS/CMOS battery photograph.